Reflections on the PID Controller project

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Elevator Pitch.
As a person who's extremely excited about embodied AI, here's some evidence that I have worked on it: (1) First Authored Published paper on World Models (ML based simulating environments for bots) at NeurIPS'26 workshop. (2) Co-president of Santa Cruz AI Club, (3) I've worked on training agents for games (Under Jim Whitehead's Lab). I am really excited about the applications of ML in terms of real world hardware, and would love to contribute as much as I can to the project

Summary of project.
I tried two approaches to make the controller for the velocity control for a car! PID Controller and a Multilayer perceptron based on synthetic data produced on the PID. The main thesis was to see how well can the MLP imitate the PID, which it was successfully executed (with small improvements).

What I've learned.
> How a basic PID works (It's a really interesting and simple control model, probably the "Hello World!" of it).
> A lot about the model trained depends on the data it's trained on.
> The MLP showed a small bit of a lesser error, so maybe non rigid formulas work better for control. A bit of noise from the MLP could be useful, as it was consistently better through different training runs.

Future Work.
Here are a few directions I was thinking about if I had more time with the project
> Working towards an RL model with reward function calculated based on the error
> Add a visual simulation for understanding the car better, maybe even train it in the environment to make it a bit more realistic (World models would be a light weight thing), I'm still kind of curious about how different the results would be in terms real life environments

Exciting points.
I am really excited to see how different software control algorithms work in terms of a RACECAR! (need I mention, working on a racecar is one of the most exciting things). I am really excited about world models (just published a paper about them in NeurIPS'26 workshop), and or agent training environments.